Testosterone, diversity, and group project performance

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1 Introduction

For many years researchers have been trying to find out the effects of diversity on the group's performance. The group performance is more relevant in the context of competition. On one hand, it has been seen that diversity can increase performance because the diverging perspectives of group members can lead to better decisions and more creative ideas and solutions. However, diverse groups can have more conflicts for eg. due to stereotyping, discrimination and hate for other races or due to feelings of superiority or inferiority as compared to other races eventually affecting the performance as explained in (Modupe 2018). Thus, Diversity can have both positive and negative effects. Similarly, homogeneous groups can perform better as they will have fewer conflicts and are more focussed towards goal. However, as explained in (Modupe 2018) homogenous groups due to lack of perspective may take inferior decisions affecting the performance.

Apart from diversity, Testosterone, a steroid hormone, is associated with motivation to attain status and hence plays an important role during individual or group competitions. As explained in (Modupe 2018), High Testosterone exhibiting dominance-related behaviors and these individuals tends to outperform others in the competition. However similar behavior in a group can lead to conflict eventually hindering the performance.

There is another hormone called cortisol. It works with certain parts of the brain to control mood, motivation, and fear. A high level of this may lead to depression in anxiety. During the competition, there is a lot of stress developed. Controlling stress and anxiety is very important in competition as it can affect the decision ability of individuals as well as groups. Cortisol helps in managing stress. Hence, we need to consider the cortisol level

In this work, we are using a dataset from an existing experiment (Modupe 2018). The study involved 370 students working on their Master of Business Administration degrees at Columbia Business School. The students were randomly assigned to groups of three to six people, and their testosterone levels were measured. The groups then did a week-long project. Finally, their performance was scored. We attempt to investigate the effect on a group's performance due to the combined effect of both hormones and diversity at the group level.

For our work we attempt to test the following hypothesis:

- H1 Diversity does not improve the performance improves
- H2 High level of average testosterone does not improve the Performance

Variable	Description
team.id	ID number of the team this participant belonged to
Age	Age, in years
Gender	Gender (Male or Female)
Ethnicity	Ethnicity of the participant
Cortisol	Participant's cortisol levels, nMol/L
Testosterone	Participant's testosterone levels, pg/mL
Country	Country of citizenship of the participant

Table 1: Individual data

- H3 High level of average cortisol does not improve the Performance

1.1 Data Summary

From the provided dataset, there are two types of data files, the first files contain information about 370 individual participants as shown in the table 1. And the second file contains information about 74 different groups that were formed from 370 participants as shown table: 2

1.2 Handling Missing data

In the given individual data, there were few members whose some of the details from table 1 like Cortisol, Testosterone, age was missing. To handle this missing data We simply remove these members from the Individual list as well as there details from the team to which they belong. We then adjusted the team's details, however, we keep the final performance to be the same. For eg. In Individual data, there is a member, ID 140 with missing Cortisol and Testosterone level. This individual's gender is Female, Ethnicity black, Age 26, Country USA, and has been assigned team 16. In Team 16 the team size is 5, the number of females 2. To adjust the effect of removed member we simply change team size to 4 and the number of females to 1. Since the number of individuals with missing details is just 18 out of total 370, we believe this way of handling missingness will not impact the conclusion.

1.3 Including preprocessed information in given data

Since our focus is on studying the team's performance we need to define scores for each team which would take into account the age, Testosterone, cortisol, gender, country ethnicity of each of its members. Hence, we define the team's age score, diversity score, Testosterone score, cortisol score and gender score as defined in table 1. The calculation is done on the cleaned dataset.

Variable	Notations	Description
team.id	tid	Team ID number
team.size	ts	Number of people on the team
final.performance	fp	The team's final performance score
females	nf	Number of females in the group
gender.score	gs	Ratio of females in the group
diversity.score	ds	Unique gender-ethnicity-country combinations
age.score	age	Team's average age
Testosterone.score	at	Team's average Testosterone level
Cortisol.score	ac	Team's average Cortisol level

Table 2: Team data

2 Exploratory Data Analysis

For the current study, the response variable is the team's performance. We first explored if there is any relationship between latent and observable variables listed in table 2.

From figure 1 we observed testosterone score decreases with gender score. Here gender score is the female ratio and the female gender usually has low testosterone level, hence the trend. Thus we have considered the relationship between gender.score and testosterone score. The cortisol level seems to increase first and decrease with gender score. On average there seems to be no relation between cortisol and gender score. The observed trend must be due to some other external factors. Hence we have not considered any relationship between them. We also did not find any relationship between diversity score and testosterone and cortisol. But age seems to affect both hormone levels. Testosterone score increases faster with age. It has been observed that in general testosterone increases till the age of 35 and then decreases. That is why we see less enthusiasm among older people. Since the average team age is between 25 -30 it justifies the trend. We also see a weak relationship between cortisol score and age score. Hence we have considered age. score to affect both hormonal scores.

Here we discuss what variables are affecting the performance. In Figure 2, in the top-left figure, with age the performance tends to decrease and then increase. With age experience increases and hence teams performance should increase since the number of points is not enough for lower age, 25-27, the decrease in performance cannot be conclusive. The performance was supposed to remain almost the same or increase. Hence we have considered a relationship between age and performance.

In the top-middle figure, the increase in diversity suggests a decrease in performance. High diversity leads to conflict hence performance can decrease. At the same time, there are cases where performance is very high with a high diversity score.

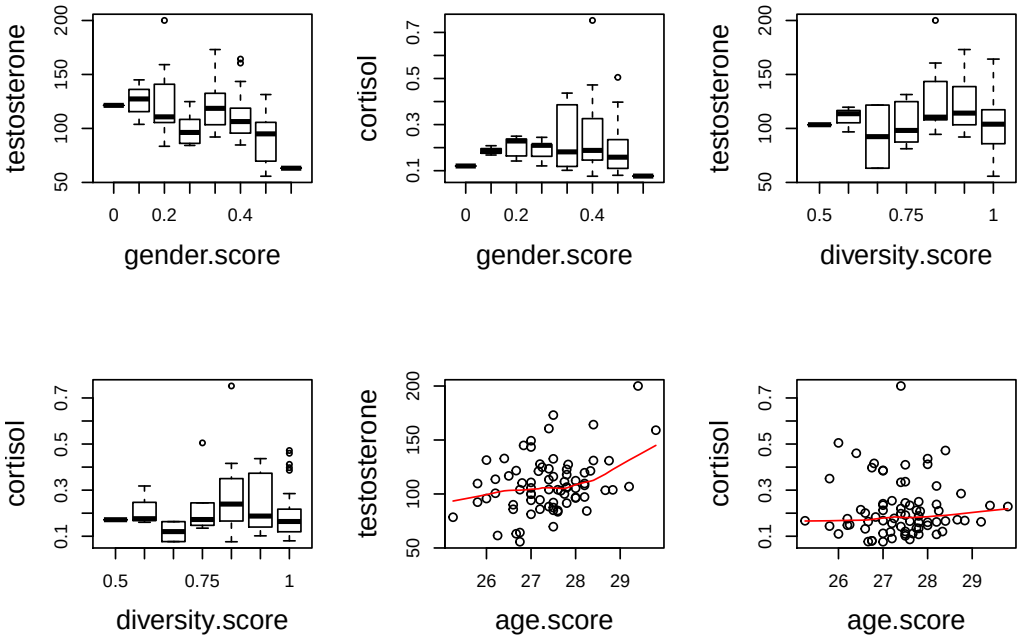

Figure 1: plots to compare non outcome variables relations

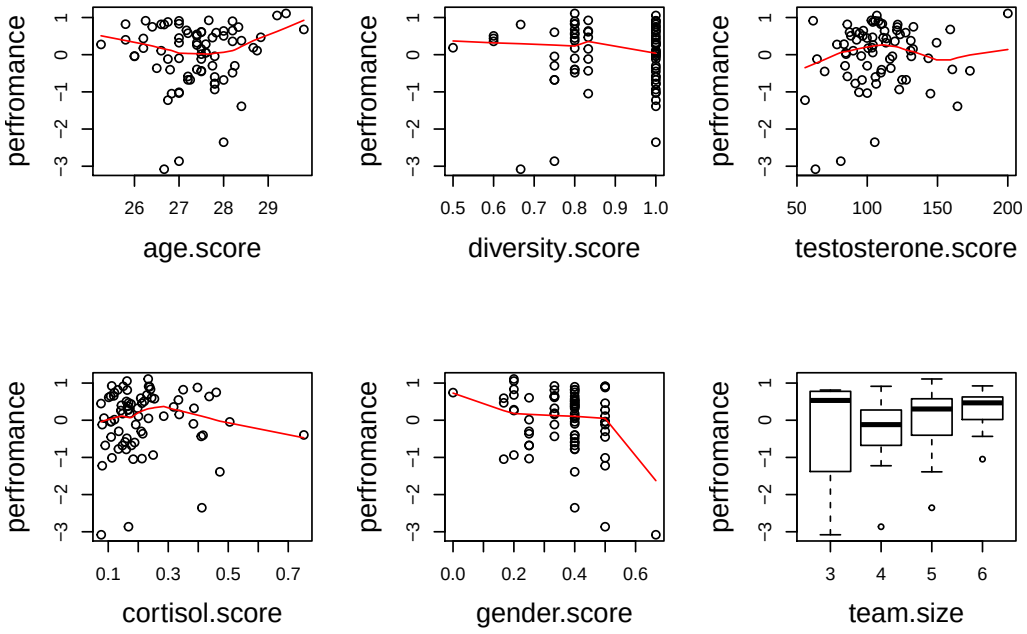

Figure 2: plots to compare team performance with other variables

This can be because there are some teams with small size suggesting no conflict and diverse perspective leading to high performance. From the bottom right figure, We also find that teams with size 3 have higher performance. This is since these teams have a very high diversity score. As the team size goes to 4, most of the team still has a very high diversity score which suggests an increase in conflict and thus reduced performance. Further, we observe that the increase in team size performance increases on average. It suggests that team size starts dominating the diversity factor. Although with very large teams and very high diversity score the performance is very mostly below average. Thus we have considered a direct relationship of diversity score with performance, team size with performance and interaction variable between team size and diversity score with performance in our model

In Top-right, If we ignore the isolated point with very high testosterone, the performance initially increases with testosterone score and then decreases. This trend suggests that initially competitive spirit in team increase but then with further increase it tends to create internal conflicts as testosterone increases the desire to dominate. Performance tends to decrease. We have also observed that small teams have very high testosterone scores. The competitiveness of smaller teams with high testosterone is very high also there less conflict due to fewer people trying to dominate. Thus we will include an interaction of testosterone and team size in our model.

Based on observation, we have also included an interaction between testosterone and diversity score as for large team size high testosterone and high diversity decreases the performance but for small teams, high testosterone and high diversity increase the performance.

From figure 1 we find that only age has a positive relationship with the cortisol level. Ignoring isolated cases in the end, the performance improves with an increase in cortisol score. Since with higher cortisol level stress is properly managed. Hence we consider a direct relationship between cortisol and stress

Finally, the performance decreases with gender score, this is because testosterone level also decreases. Hence we do not consider gender score to have any direct effect on performance.

3 Method

Based on prior information and exploratory data analysis we design our causal diagram as shown in figure:3.

Based on causal model we test the model as shown in following equation: For now, we use a short form of each variable in our equations as follows: ts-team.size, ds-diversity.score, lat-log.testosterone.score, lac-log.cortisol.score, age-age.score, tsds-interaction of teamsize and diversity.score.

$$fp = f(ts, ds, lat, lac, age, tsds) \tag{1}$$

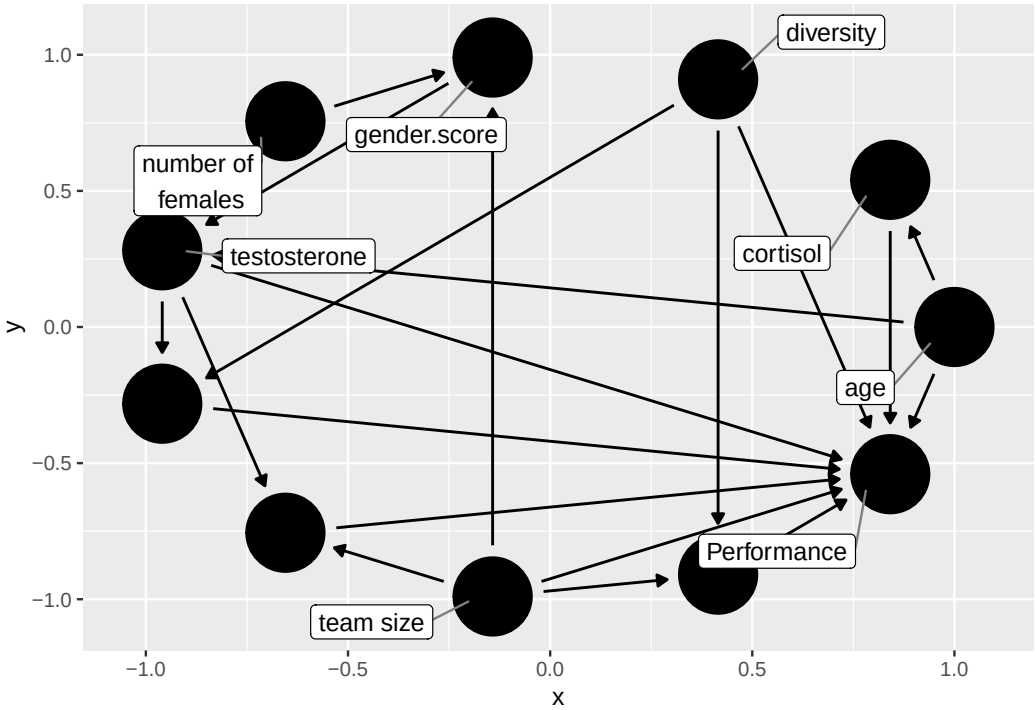

Figure 3: plots to compare variables relations

We use the linear regression model to test our hypothesis.

Table 3: Coefficient for model

	Estimate	Pr(>|t|)
(Intercept)	-82.98	0.00
ts	9.13	0.00
ds	46.66	0.03
lat	17.22	0.00
lac	-0.14	0.52
age	-0.04	0.76
ts:ds	-0.64	0.47
ts:lat	-1.79	0.00
ds:lat	-9.41	0.03

In linear regression model, the coefficient represents, by how much performance will increase if the corresponding variable is increased by the value of one. From the table 3 we observe that

- Using t-test the p-value for team size, log testosterone and diversity score, the interaction of team size and testosterone and interaction of testosterone and diversity score to be significant.
- There is a positive relationship of performance with diversity. the score, log.tesotsterone.score and team size.
- The coefficient of interaction between and diversity.score and log.testosterone.score is negative.
- The coefficient of interaction between and team.size and log.testosterone.score is negative.
- The coefficient of cortisol.score is very less and in fact, its p-value is not significant.

3.1 Diagnostics of Linear model

From the diagnostics in figure 4

- QQ plot is almost along the line, which says that the distribution of data is almost normal, except three points
- Residual vs fitted value is almost horizontal, which shows that the model we get is close to linear, although point 58 affects the model a lot
- The scale graph is not linear it means the data is not homogeneous
- The graph of residual vs Leverage has some weird patterns, it may be due to point 58 affecting the regression since apart from team 58 and team 2 rest, other teams, performance score is more than -.5

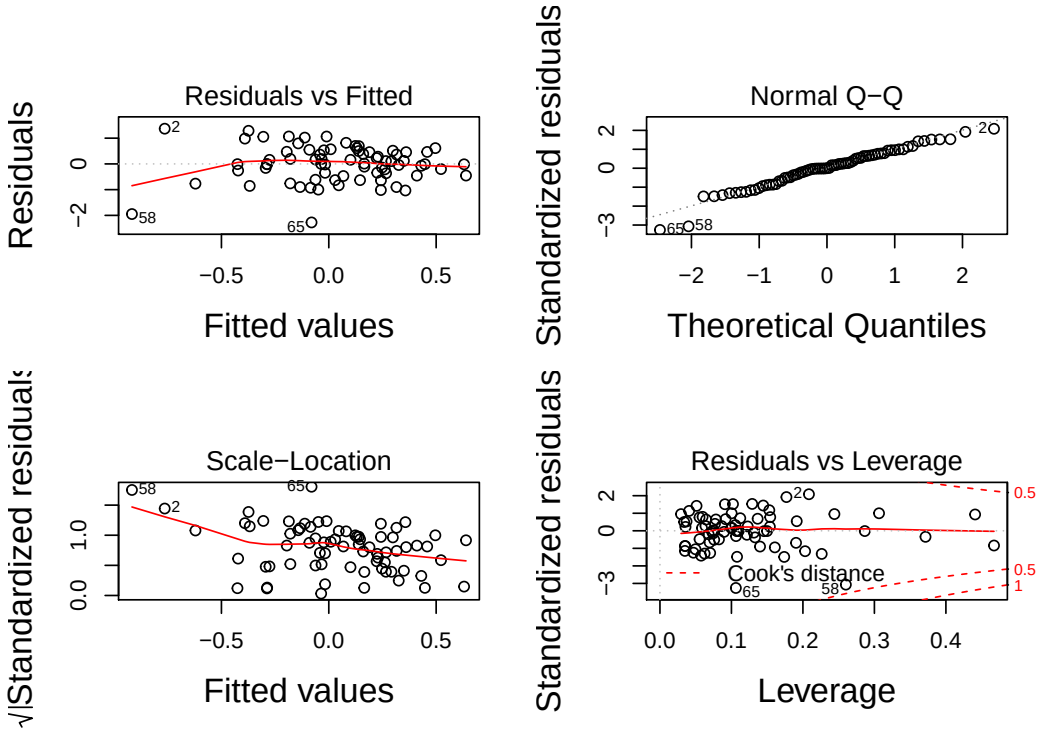

Figure 4: Diagnostics

4 Conclusion

With the near satisfaction of assumptions of linear regression, we find our learned model to be almost accurate. Age and Gender play an important role in performance. With Age from 25-35, the testosterone level increases with age whereas the female has a lower testosterone level. And, from the coefficient value, we know how important is the testosterone level for performance. gender score and age does not affect the performance directly but through testosterone level.

Diversity score also plays a very important role in improving performance. However, for large team size, the high testosterone level is not good for performance. Hence, In very large groups for eg. it is good to include females which will increase diversity score and reduce the overall testosterone level. Higher diversity can lead to conflict and so is the higher testosterone, based on the interaction coefficient of diversity and testosterone, for higher diversity, it is better to have low testosterone and for lower diversity, it is better to have high testosterone level in the team.

From table 3, based on the coefficient values corresponding p-value we conclude that forgiven team size diversity score and testosterone score improves the team's performance while cortisol does not affect the performance. Hence, we reject null hypothesis H1, reject null hypothesis H2 and accept null hypothesis H3

Finally, from diagnostic plots, there is some scope of improving the model. We have considered diversity score as a unique combination of gender-ethnicity-country. However, this definition is too simple. It does not take into account the real effect of ethnicity or country. For eg., A person coming from any west European country may think similarly. Hence, we need a better definition of diversity score. The data is not homogeneous. variance correction needs to be done before running the model

5 Reference

Modupe, Akinola. 2018. "Hormone-Diversity Fit, a Collective Testosterone Moderates the Effect of Diversity on Group Performance" 29 (6): 859–67. https://doi.org/10.1177/0956797617744282.